FRED-MD | Data Cleaning & Transformation Guide

Project handoff • R / Databricks • Source vintage: September 2026

1. Scope and data source

The September 2026 vintage of FRED-MD was imported into Databricks from a CSV. The source contains 812 monthly observations (January 1959–August 2026) and 126 economic indicators. All economic predictors come from FRED-MD; the NBER recession target is not included in these deliverables. The raw source table is **hive_metastore.default.fred_md_202609**.

2. Processing steps

• Preserved the first CSV data row labeled Transform: as a named vector of 126 transformation codes; removed that row from the monthly observations.

• Converted sasdate to an R Date and all economic series to numeric; sorted observations chronologically.

• Applied the FRED-MD code for each series individually (codes 1–7), preserving unavailable values as NA.

• Verified the 812-month date range, no duplicate dates, no missing calendar months, and no infinite transformed values.

• Examined missing values by series and date, including the October 2025 raw-data gap and its effects on later transformed months.

• Created provisional macroeconomic and financial predictor groups (100 and 26 indicators, respectively).

• Exported CSV files and saved corresponding Databricks tables; checked row and column counts and the full transformed table's date range and NA count.

3. FRED-MD transformation codes

Code	Operation
1	Level: x(t)
2	First difference: x(t) − x(t−1)
3	Second difference: $\Delta^2 x(t)$
4	Natural log: ln x(t)
5	First log difference: $\Delta \ln x(t)$
6	Second log difference: $\Delta^2 \ln x(t)$
7	Difference in simple growth rates: $\Delta[x(t)/x(t-1) - 1]$

Log transformations use natural logarithms; log differences were not multiplied by 100. The assigned code is series-specific, not inferred from macro/financial group membership. Example checks: RPI=5, UNRATE=2, FEDFUNDS=2, CPIAUCSL=6, T10YFFM=1 and VIXCLSx=1.

4. Missing-data treatment

The transformed full dataset retains **1,209 NA cells** across 812 × 126 predictor cells; 407 months have complete predictors. No imputation or complete-case deletion was performed. Early missing values often reflect limited series coverage or differencing. October 2025 has 21 missing raw predictors; these create 21 missing transformed predictors in November and 10 in December for lag-dependent transformations. Imputation and scaling must be fit only on training data by the modeling team.

5. Deliverables and verified dimensions

Databricks table (default schema)	Rows	Columns
fred_md_macro_clean	812	101

fred_md_macro_financial_clean	812	127
fred_md_transformed_clean	812	127
fred_md_variable_dictionary	126	5

The dictionary includes Variable, Group, Transformation_Code, Missing_Count and Missing_Percent. Delta-compatible column names may replace spaces and special characters with underscores (e.g., S&P 500 → S_P_500). Keep a mapping to the original FRED-MD names when sharing.

6. Limitations and teammate handoff

The 100/26 split is a **provisional research classification**, not an official FRED-MD grouping. Money and credit variables currently classified as macroeconomic should be reviewed by the team. All 812 dates are retained in the master files, but the number of usable labeled observations may be smaller once forecast horizons and NBER targets are constructed. The September 2026 vintage includes historically revised values and is not a fully real-time information set.

Recommended use: preserve the original raw file and this guide; use the transformed tables as predictor inputs; align any recession target by month; avoid full-sample imputation or scaling; use the same evaluation dates when comparing macro-only and macro-plus-financial specifications.

Source: Federal Reserve Bank of St. Louis, FRED-MD database (McCracken and Ng), September 2026 vintage; transformation codes supplied in the source file.